

A Cartesian Closed Fibration of Higher-Order Recognizable Languages

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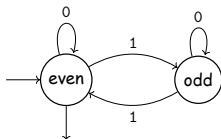
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Higher-order automata theory

There is a growing connection between

automata theory



and

λ -calculus

$\text{Words}_\Sigma \Rightarrow \text{Bool}$

centered around higher-order languages, as defined by Salvati using recognition by FinSet . Today's presentation is about

the cartesian closed and fibrational structure
of higher-order languages.

Fundamental closure properties

Recognizable languages of finite words are stable under

- Boolean operations:

$$L \cap K, \quad \Sigma^*, \quad L \cup K, \quad \emptyset, \quad \complement L$$

- Residuals:

$$a \setminus L = \{w \in \Sigma^* \mid aw \in L\}, \quad L/a = \{w \in \Sigma^* \mid wa \in L\}$$

- Inverse image:

$$u^{-1}(L) = \{w \in \Sigma^* \mid u(w) \in L\} \quad \text{along homomorphisms}$$

The category of recognizable languages

We consider the category whose objects are pairs

$$(\Sigma, L) \quad \text{for } L \in \mathbf{Rec}(\Sigma)$$

and whose morphisms

$$(\Sigma, L) \longrightarrow (\Gamma, K)$$

are monoid homomorphisms

$$u : \Sigma^* \longrightarrow \Gamma^*$$

such that for all $w \in L$, we have $u(w) \in K$.

$$\begin{array}{ccc} & (\Sigma, L') & \\ \uparrow \text{Id}_{\Sigma^*} & & \\ (\Sigma, L) & \xrightarrow{u} & (\Gamma, K) \\ \downarrow & & \downarrow \\ \Sigma^* & \xrightarrow{u} & \Gamma^* \end{array}$$

Direct images

In the case of finite trees, recognizable languages are preserved by direct images of homomorphisms that are linear.

In a calculus of recognizable languages of tuples of finite words,

$$\begin{array}{ccccc}
 \mathbb{N} & \xrightarrow{\text{direct image}} & \{(n, n) : n \in \mathbb{N}\} & \xleftarrow{\text{inverse image}} & \{w \in \Sigma^* \mid |w|_a = |w|_b\} \\
 \cap & & \cap & & \cap \\
 \mathbb{N} & \xrightarrow{n \mapsto (n, n)} & \mathbb{N} \times \mathbb{N} & \xleftarrow{(|w|_a, |w|_b) \leftarrow w} & \Sigma^*
 \end{array}$$

so inverse and direct images cannot always coexist.

From words to λ -terms

Every finite alphabet $\Sigma = \{a_1, \dots, a_n\}$ induces a simple type of the λ -calculus

$$\text{Words}_\Sigma = \underbrace{(\mathbb{O} \Rightarrow \mathbb{O})}_{a_1} \Rightarrow \dots \Rightarrow \underbrace{(\mathbb{O} \Rightarrow \mathbb{O})}_{a_n} \Rightarrow \underbrace{\mathbb{O}}_{\text{input}} \Rightarrow \underbrace{\mathbb{O}}_{\text{output}}$$

and every finite word $w = a_{i_1} \dots a_{i_l}$ induces a λ -term

$$\lambda a_1. \dots \lambda a_n. \lambda x. a_{i_l} (\dots (a_{i_1} x))$$

This yields a bijection, called **Church encoding**

$$\Sigma^* \cong \{\text{closed } \lambda\text{-terms of type } \text{Words}_\Sigma\} / \beta\eta$$

Higher-order recognition

For any finite set Q interpreting the base type $\llbracket \circ \rrbracket_Q := Q$, we associate

$$M : A \quad \rightsquigarrow \quad \llbracket M \rrbracket_Q \in \llbracket A \rrbracket_Q$$

A **higher-order language** L is one recognized by **FinSet**, i.e. of the form

$$L = \{M : A \mid \llbracket M \rrbracket_Q \in F\} \quad \text{for some finite } Q \text{ and } F \subseteq \llbracket A \rrbracket_Q$$

Higher-order languages of λ -terms of type A form a

Boolean algebra $\mathbf{Rec}(A)$

We get back recognizable languages of finite words when $A = \mathbf{Words}_\Sigma$.

Product and arrow languages

For any types A and B , we can form the types

$$A \times B \quad \text{and} \quad A \Rightarrow B$$

which have universal properties in the category **Lam** of types and λ -terms.

For any languages L and K of λ -terms of type A and B respectively, we define

$$L \times K \quad := \quad \{(M, N) : A \times B \mid M \in L \text{ and } N \in K\}$$

$$L \Rightarrow K \quad := \quad \{F : A \Rightarrow B \mid \text{for all } M \in L, \text{ we have } FM \in K\}$$

of λ -terms of type $A \times B$ and $A \Rightarrow B$.

The category of higher-order languages

The category **Rec** has as objects the pairs

$$(A, L) \quad \text{for } L \in \mathbf{Rec}(A)$$

and as morphisms

$$(A, L) \longrightarrow (B, K)$$

the closed λ -terms

$$F : A \Rightarrow B$$

such that for all $M \in L$, we have $FM \in K$.

$$(A, L')$$

$$\uparrow \text{Id}_A$$

$$(A, L) \xrightarrow{F} (B, K)$$

$$\downarrow$$

$$\downarrow$$

$$A \xrightarrow{F} B$$

We write $p : \mathbf{Rec} \rightarrow \mathbf{Lam}$ the forgetful functor $(A, L) \mapsto A$.

Main theorems

Theorem A. Higher-order languages are stable under $\times$ and $\Rightarrow$.

Theorem B. Higher-order languages are stable by inverse image of λ -terms.

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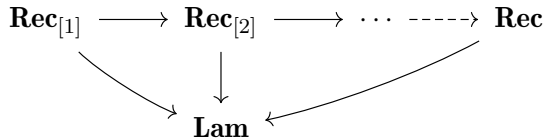
Theorem A+B. The functor $p : \mathbf{Rec} \rightarrow \mathbf{Lam}$ is a **cartesian closed fibration**.

We give two proofs of this theorem:

1. one using Frobenius reciprocity to glue countably many fibrations
2. one using **profinite methods**, which uses a single change of base

One word on the first proof

We build a colimiting diagram of cartesian closed fibrations



using a **Frobenius reciprocity** condition to get cartesian closed functors.

Stone duality

A Stone space is a compact space that is totally separated.¹

They are dual to Boolean algebras:

$$\mathbf{Stone} \cong \mathbf{BA}^{\text{op}}$$

For any type A , the space of profinite λ -terms of type A

$$\widehat{\Lambda}(A) \quad \text{is the Stone dual of} \quad \mathbf{Rec}(A)$$

These spaces assemble into a functor $\widehat{\Lambda} : \mathbf{Lam} \rightarrow \mathbf{Stone}$.

¹a separation axiom with clopens, stronger than being Hausdorff.

The clopen fibration

The category Clopen has as objects the pairs

(X, U) of the Stone space X and a clopen $U \subseteq X$

and as morphisms $(X, U) \rightarrow (Y, V)$ the continuous functions

$$f : X \longrightarrow Y$$

such that for all $x \in U$, we have $f(x) \in V$.

Clopen subsets

$$U \subseteq X$$

correspond to
continuous
functions

$$X \longrightarrow 2$$

Theorem (Stone duality). The forgetful functor $\text{Clopen} \rightarrow \text{Stone}$, defined as $(X, U) \mapsto X$, classifies fibrations of Boolean algebras.

Higher-order languages via classification

By Stone duality, we get the fibration $\mathbf{Rec} \rightarrow \mathbf{Lam}$ in a single change of base:

$$\begin{array}{ccc} \mathbf{Rec} & \longrightarrow & \mathbf{Clopen} \\ \downarrow & \lrcorner & \downarrow \\ \mathbf{Lam} & \xrightarrow{\hat{\Lambda}} & \mathbf{Stone} \end{array}$$

Moreover, we observe that

- ▶ The functor $\hat{\Lambda} : \mathbf{Lam} \rightarrow \mathbf{Stone}$ preserves cartesian products
- ▶ The fibration $\mathbf{Clopen} \rightarrow \mathbf{Stone}$ has simple products

Theorem. $p : \mathbf{Rec} \rightarrow \mathbf{Lam}$ is a cartesian closed functor.

An Isbell-type adjunction for languages

Using the structure of cartesian closed fibration, we get:

Theorem. For any λ -term

$$M : A \times B \longrightarrow C$$

and language $L \in \mathbf{Rec}(C)$, we have an adjunction

$$\mathbf{Rec}(A) \begin{array}{c} \xrightarrow{(-) \setminus L} \\ \perp \\ \xleftarrow{L / (-)} \end{array} \mathbf{Rec}(B)^{\mathrm{op}}$$

where for every $K \in \mathbf{Rec}(A)$, we have

$$K \setminus L = \{ N_B : B \mid \text{for all } N_A \in K, \text{ we have } M(N_A, N_B) \in L \}$$

Reconstructing the residuals

Fact. Taking the types $A = B = C = \text{Words}_\Sigma$, the λ -term

$$M := \text{concat} \quad : \quad \text{Words}_\Sigma \times \text{Words}_\Sigma \longrightarrow \text{Words}_\Sigma$$

and any language $L \in \mathbf{Rec}(\Sigma)$, we recover the residuals

$$\mathbf{Rec}(\Sigma) \begin{array}{c} \xrightarrow{(-)\backslash L} \\ \perp \\ \xleftarrow{L/(-)} \end{array} \mathbf{Rec}(\Sigma)^{\text{op}}$$

computed on regular expressions by **Brzowski derivatives**.

Open λ -terms

A λ -term $M: A \Rightarrow B$, preserves the higher-order languages by direct image

$$\exists_M : \mathbf{Rec}(A) \longrightarrow \mathbf{Rec}(B)$$

if and only if the continuous function

$$\hat{\Lambda}(M) : \hat{\Lambda}(A) \longrightarrow \hat{\Lambda}(B)$$

is open for the profinite topologies.

In that case, we say that M is an **open** λ -term.

Examples and non-examples

The following λ -terms are open

M	:	$1 \longrightarrow A$	due to Statman's finiteness theorem
π	:	$A \times B \longrightarrow B$	induces quantifiers of languages
$\underline{u}$	:	$\text{Words}_\Sigma \longrightarrow \text{Words}_\Gamma$	for any homomorphism $u : \Sigma^* \rightarrow \Gamma^*$

The following λ -terms are not open in general

$$\delta : A \longrightarrow A \times A \quad \text{and} \quad \text{ev} : (A \Rightarrow B) \times A \longrightarrow B$$

- Lawvere's equality predicate is not regular

Conclusion

Future work:

- ▶ Characterization of open λ -terms
- ▶ Higher-order languages of linear λ -terms
- ▶ Higher-order parity automata and infinitary λ -terms
- ▶ Towards higher-order Kleene stars and regular expressions
- ▶ MSO properties and higher-order transducers

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Thank you for your attention!

Any questions?